GRADUATE INSTITUTE OF MEDICINE • I-SHOU UNIVERSITY

Design of Clinical Trials

Principles III

Research hypotheses and study endpoints



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Four goals for this session



Formulate

Turn a clinical problem into a precise, testable hypothesis with explicit H_0 and H_1.



Select

Explain the roles of primary, secondary, and exploratory endpoints and choose an appropriate one.



Manage

Handle multiple endpoints: multiplicity adjustment, hierarchical testing, composite and co-primary endpoints.



Apply

Apply good practice: prespecification, clinical relevance, and transparent documentation.

A hypothesis, its endpoint, and its analysis are designed together, not one after another.

From clinical problem to testable hypothesis

The question narrows at each step; the hypothesis is the last and most precise statement.



Clinical problem	Literature review	Objective(s)	Hypothesis	Endpoint
Broad uncertainty seen in practice	Identify the knowledge gap	Primary and secondary aims	H_0 versus H_1, measurable	What is measured, when, how

Broad → Narrow

Example "Does hydroxychloroquine help in COVID-19?" → "Among adults within 4 days of a high-risk exposure, does hydroxychloroquine reduce symptomatic, laboratory-confirmed COVID-19 by day 14 compared with placebo?"

A hypothesis you cannot measure is a wish, not a design.

PICOT: the anatomy of an answerable question

Every element of the question reappears later as an element of the design.

	Element	Design content	NCT04308668 example
P	**Population**	Who is studied; eligibility criteria	Adults ≤4 days after household or occupational exposure to SARS-CoV-2
I	**Intervention**	The test treatment, dose, duration	Hydroxychloroquine, 5-day course
C	**Comparator**	Placebo, active control, standard care	Matching placebo (folate)
O	**Outcome**	The endpoint and its measurement	Symptomatic illness compatible with COVID-19 or PCR-confirmed infection
T	**Time**	When the outcome is assessed	Within 14 days of enrollment

If one PICOT element is vague, the hypothesis, the endpoint, or the sample size will be vague too.

Null and alternative hypotheses

The trial is designed to reject H_0; it never proves H_1 and never proves H_0.

H_0 Null hypothesis

The treatment has no effect (or no effect beyond the margin). It is the position the data must argue against.

Example H_0: incidence of COVID-19 with HCQ = incidence with placebo

H_1 Alternative hypothesis

The treatment has an effect. It is what the investigator hopes to demonstrate, and it determines the sample size.

Example H_1: incidence with HCQ ≠ (or <) incidence with placebo

Link to Principles II

Claim	H_0	H_1	Evidence required
Superiority	$d \leq 0$	$d > 0$	Lower CI bound > 0
Non-inferiority	$d \leq -\Delta$	$d > -\Delta$	Lower CI bound > $-\Delta$
Equivalence	$d \leq -\Delta$ or $d \geq +\Delta$	$-\Delta < d < +\Delta$	Entire CI within $(-\Delta, +\Delta)$

Write the hypothesis on the scale of the endpoint, with the direction of benefit stated explicitly.

What makes a hypothesis testable

Practical checks before the protocol is written

Specific

One population, one comparison, one primary outcome, one time point.

Measurable

The outcome can be observed with a defined instrument and a defined rule for success.

Directional

States whether the aim is superiority, non-inferiority, or equivalence, with any margin.

Feasible

Event rates, recruitment, follow-up, and budget allow the required sample size.

Ethical and lawful

Clinical equipoise exists; the design meets regulatory and IRB expectations.

Documented

Written in the protocol and statistical analysis plan, and registered before enrollment.

Statistical input belongs at the hypothesis stage, not after the data arrive.

Common pitfalls and their fixes

Pitfall	Fix
Overly broad *"Drug X improves outcomes in liver disease."*	**Specific** *"Drug X reduces 48-week fibrosis progression in compensated MASH cirrhosis."*
Ambiguous language *"Patients feel better on treatment."*	**Clear** *"Mean change in the FACIT-F fatigue score at week 24 is greater with treatment."*
Unfeasible objective *"Drug X lowers 20-year mortality."*	**Reasonable** *"Drug X lowers 3-year liver-related events, a validated surrogate pathway."*
Ethically questionable *"Placebo for patients with an effective standard of care."*	**Lawful** *"Add-on to standard care versus standard care alone."*

Each weak hypothesis can be repaired by adding the missing PICOT element.

CLINICAL EXAMPLE

Hydroxychloroquine post-exposure prophylaxis

NCT04308668: a randomized, double-blind, placebo-controlled trial

"We hypothesized that hydroxychloroquine could potentially be used as postexposure prophylaxis, to prevent symptomatic infection after exposure to Covid-19."

Objective 1

Post-exposure prophylaxis

Prevent symptomatic COVID-19 after a known high-risk exposure

Primary endpoint: incidence of laboratory-confirmed COVID-19 or compatible illness within 14 days

Objective 2

Early preemptive therapy

Prevent disease progression in persons already symptomatic

Endpoints: hospitalization, symptom severity (0–10 visual analog scale) over 14 days

Two objectives, two hypotheses, two endpoints: each hypothesis has its own measurable outcome.

Boulware DR et al. N Engl J Med 2020;383:517–525

An endpoint is the hypothesis made measurable

A specific outcome, measured in a defined way at a defined time, used to judge the effect of the intervention



Clinical relevance

Reflects how a patient feels, functions, or survives, or a validated surrogate of it.

Measurability

Objective, reproducible, with a validated instrument and a prespecified rule for success.

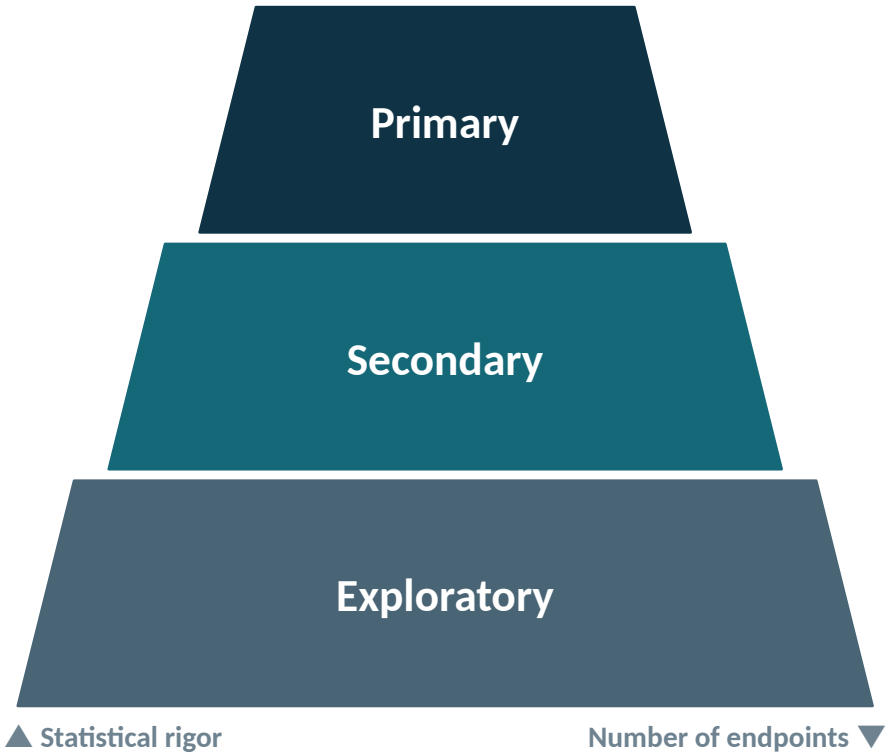
Feasibility and acceptability

Can be observed in the available time and population, and is acceptable to patients and regulators.

The primary endpoint fixes the analysis and the sample size; change it and the whole design changes.

Three tiers of endpoints

The tier determines the statistical rigor, not the clinical interest



Primary

The main outcome of interest. Basis for the hypothesis test, the success criterion, and the sample size. Usually one.

Secondary

Prespecified additional outcomes: support the primary result, describe other effects, or characterize safety. Usually several; formal claims need multiplicity control.

Exploratory

Hypothesis-generating. Not powered, flexible, may be post hoc. Findings are preliminary and not a basis for regulatory claims.

Only the primary endpoint carries the confirmatory claim; the others support, explain, or generate hypotheses.

Primary, secondary, and exploratory endpoints compared

Aspect	Primary endpoint	Secondary endpoints	Exploratory endpoints
Purpose	Test the main hypothesis; answer the primary objective	Evaluate additional prespecified effects	Generate new hypotheses; explore further questions
Statistical power	Trial is powered to detect the targeted difference	May be powered, but usually to a lesser extent	Not powered; analyses are often descriptive
Prespecification	Clearly defined before the trial begins	Defined before the trial; analyses planned	May be prespecified or identified during analysis
Regulatory impact	Central to approval and labeling claims	May support claims; provide additional information	Generally not used for regulatory decisions
Multiplicity	Adjustment required if more than one	Adjustment needed for formal claims	Typically no adjustment; findings are preliminary

Secondary endpoints can support a label claim only if they are prespecified and multiplicity is controlled.

CLINICAL EXAMPLE

Registered outcome measures: NCT04308668

As listed on ClinicalTrials.gov, with time frame

Primary outcome measures

Outcome measure	Description	Time frame
COVID-19 disease at day 14 (asymptomatic at baseline)	Participants with active, confirmed or compatible disease	14 days
Change in disease severity (symptomatic at baseline)	Visual analog scale 0–10 for overall symptom severity	Baseline and 14 days

Secondary outcome measures

Outcome measure	Description	Time frame
Hospitalization	Participants requiring hospitalization for COVID-19	14 days
Death	Participants who die of COVID-19-related disease	About 30 days
Confirmed SARS-CoV-2 detection	Participants with confirmed infection	14 days
Medication discontinuation	Participants who stop study medication for any reason	14 days

Registration before enrollment makes the primary endpoint publicly verifiable.

ClinicalTrials.gov NCT04308668; Boulware DR et al. N Engl J Med 2020;383:517–525

The endpoint's data type decides the effect measure

Link to Principles II: the hypothesis is written on the scale of the effect measure

Data type	Typical endpoints	Summary per arm	Effect measure	Sample size driver
Binary	Response, cure, infection, death by a fixed time	Proportion	Risk difference, risk ratio, odds ratio	Control event rate
Continuous	Blood pressure, HbA1c, score change from baseline	Mean (SD)	Mean difference	Standard deviation
Time-to-event	Overall survival, progression, first hospitalization	Median, survival curve	Hazard ratio	Number of events
Count or ordinal	Exacerbations per year, modified Rankin scale	Rate, distribution	Rate ratio, common odds ratio	Dispersion, distribution
Patient-reported	Symptoms, fatigue, quality of life	Mean score, responder %	Mean difference, responder ratio	SD, minimal important difference

Choose the data type first; the test, the margin, and the sample size formula follow from it.

Clinical outcomes versus surrogate endpoints

A surrogate substitutes for a clinical outcome; it is only as good as its validation

Clinical (“hard”) outcome

- How a patient feels, functions, or survives
- Directly meaningful; rarely disputed
- Often infrequent and late: large trials, long follow-up

Surrogate endpoint

- Biomarker or intermediate outcome on the causal path
- Earlier, more frequent, cheaper: smaller and shorter trials
- Risk: treatment effect on surrogate may not translate into benefit

Surrogate	Clinical outcome	Validation status
LDL cholesterol	Myocardial infarction, cardiovascular death	Accepted for statins and PCSK9 inhibitors
HBV DNA suppression; HCV sustained virologic response	Cirrhosis, hepatocellular carcinoma, liver death	Accepted; supported by long-term cohorts
Suppression of ventricular ectopy (CAST)	Sudden cardiac death	Failed: encainide and flecainide increased mortality

A surrogate must predict the clinical benefit, not merely correlate with disease severity.

Fleming TR, DeMets DL. Ann Intern Med 1996;125:605–613; CAST Investigators. N Engl J Med 1989;321:406–412

Measuring the endpoint well

Practical considerations that protect the validity of the comparison

Validated instruments

Use tools with established reliability, validity, and responsiveness, such as PASI, MELD, or FACIT-F.

Standard procedures

Same timing, device, laboratory, and reader across arms and sites; central reading where feasible.

Blinded adjudication

Independent committee classifies events (for example, cardiovascular death) without knowledge of allocation.

Patient-reported outcomes

Capture symptoms and function from the patient; need validated questionnaires and a defined responder threshold.

Planned adaptation

Allow for intercurrent events: dropout, rescue therapy, death before assessment. Define handling in advance.

Data integrity

Audit trail, source verification, and missing-data strategy specified in the protocol.

An endpoint is only as credible as the procedure that measures it.

The estimand: what exactly is being estimated

ICH E9(R1) asks the protocol to define five attributes, not just the endpoint

1 Treatment

The regimen being compared, including background therapy

2 Population

Patients targeted by the clinical question

3 Variable

The endpoint obtained for each patient

4 Intercurrent events

Rescue medication, discontinuation, death: how each is handled

5 Summary measure

The population-level comparison, e.g., hazard ratio or risk difference

Strategies for intercurrent events

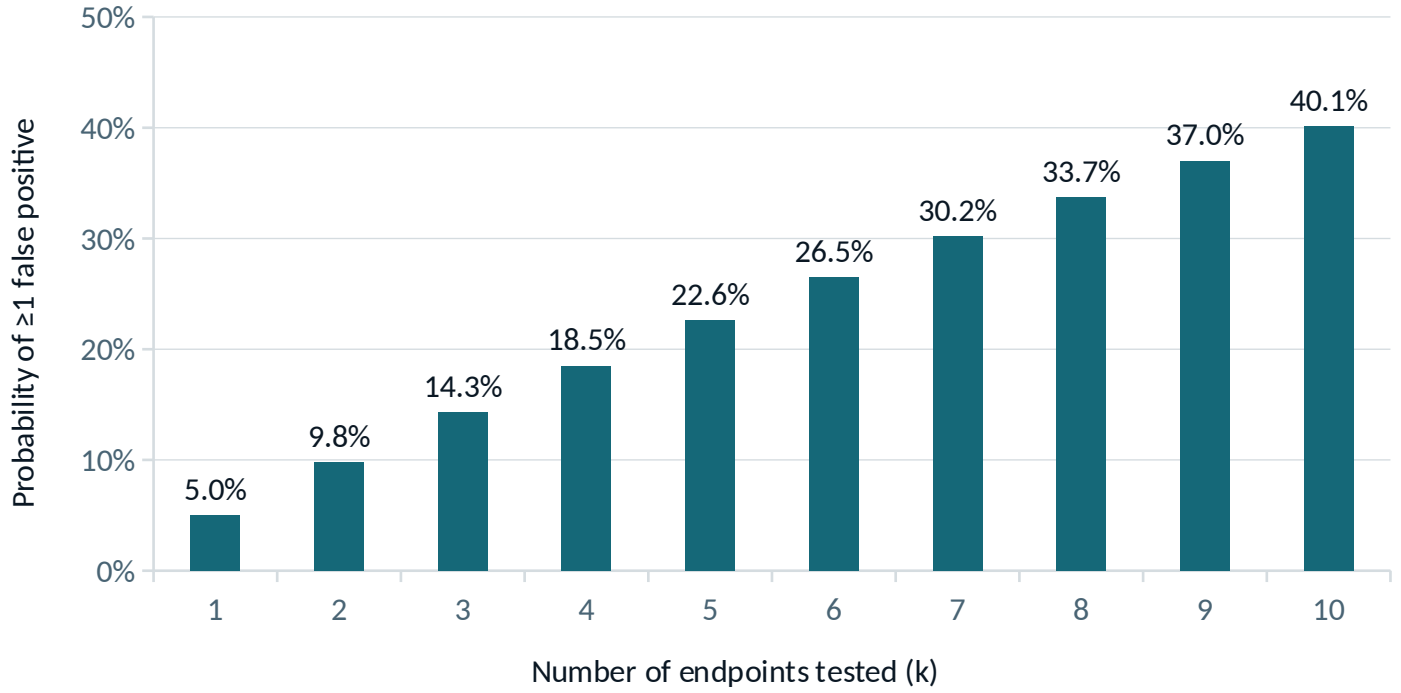
Strategy	Question answered	Example
Treatment policy	Effect regardless of what happened after randomization	Intention-to-treat analysis of mortality
Hypothetical	Effect had the event not occurred	HbA1c change as if no rescue medication had been given
Composite	Event itself counts as a (bad) outcome	Rescue therapy counted as treatment failure

Two trials with the same endpoint can estimate different things if they handle intercurrent events differently.

ICH E9(R1) Addendum on estimands and sensitivity analysis in clinical trials (2019)

Why multiple endpoints are a problem

Family-wise error rate with k independent tests, each at α = 0.05 and all null hypotheses true



k = 1

5%

k = 5

23%

k = 10

40%

Each extra endpoint inflates the chance of a false claim unless the testing plan controls it.

With five unadjusted endpoints there is roughly a one-in-four chance of at least one false-positive claim.

Family-wise error rate = $1 - (1 - \alpha)^k$, assuming independent tests

When a trial needs more than one primary endpoint

When to consider

Complex diseases

Benefit has several dimensions, e.g., histology and symptoms in MASH

Regulatory requirements

Agencies may require evidence on two domains, e.g., cognition and function in Alzheimer disease

Comprehensive evaluation

Efficacy and durability, or efficacy and safety, both matter for the decision

Challenges

Multiplicity

Type I error inflates with every additional test

Statistical complexity

Correlated endpoints, alpha allocation, power for the combination

Interpretation

What does the trial conclude if one endpoint succeeds and the other fails?

Multiplicity is a cost to be planned for, not a reason to avoid clinically necessary endpoints.

Four strategies for multiple endpoints

All must be written into the protocol and the statistical analysis plan before unblinding

Multiplicity adjustment

Claim: at least one

Split or adjust α across endpoints (Bonferroni, Holm, Hochberg). Trial succeeds if any endpoint passes its adjusted threshold.

Hierarchical testing

Claim: ordered sequence

Test endpoints in a prespecified order at full α; stop at the first failure. No alpha is spent on later tests.

Composite endpoint

Claim: single combined outcome

Combine several events into one outcome (first occurrence of any). One test, more events, higher power.

Co-primary endpoints

Claim: all

All endpoints must be significant, each at full α. No adjustment needed, but power falls and sample size rises.

The strategy is chosen by the clinical question: at least one success, all successes, or a single combined outcome.

Adjusting for multiple comparisons

Worked example: four hypotheses, overall α = 0.05

Bonferroni

$\alpha_{adj} = \alpha / m$ **for every test**

Simple and conservative; power falls quickly as m grows.

Holm (step-down)

$\alpha_{(k)} = \alpha / (m - k + 1)$ **k = rank of the p-value**

Order p-values from smallest to largest; test each against its threshold; stop at the first p-value that fails. Uniformly more powerful than Bonferroni.

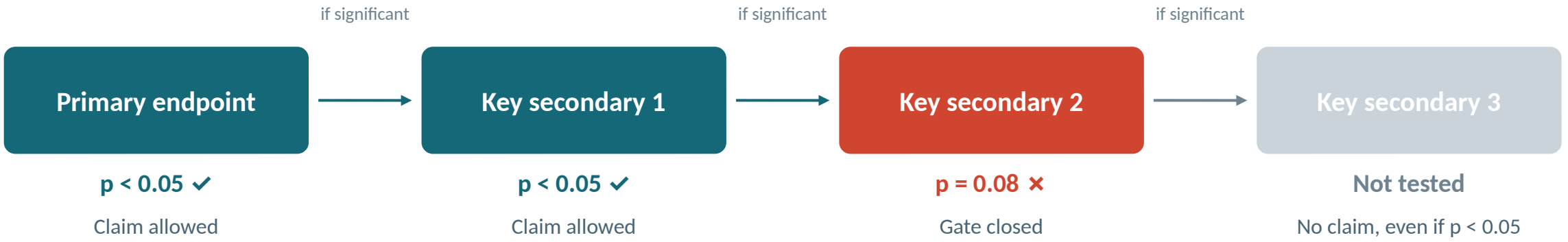
Rank	Hypothesis	p-value	Bonferroni	Holm	Decision
1	H_4	0.005	0.0125	0.0125	Reject
2	H_1	0.010	0.0125	0.0167	Reject
3	H_3	0.030	0.0125	0.0250	Stop
4	H_2	0.040	0.0125	0.0500	Not tested

Thresholds: Bonferroni α/4 for every test; Holm α/(4 − k + 1) by rank. Unadjusted testing would have declared all four significant at 0.05. Both methods reject H_4 and H_1 only; Holm's later thresholds are more generous, but H_3 still fails and testing stops.

Adjustment protects the Type I error but lowers power; the smallest p-values must clear a harder bar.

Hierarchical (gatekeeping) testing

Each endpoint is tested at the full α, but only if every endpoint before it succeeded



Strengths

- Full α at every step; no power loss for the primary endpoint
- Natural fit for a primary followed by key secondary claims
- Common in regulatory submissions

Limitations

- Order must be fixed before unblinding and cannot be changed
- A late endpoint with a strong effect yields no claim if an earlier one fails
- Variants (fallback, graphical approaches) can recycle α

Order the hierarchy by clinical importance and by the probability of success; a failure early closes every later gate.

CLINICAL EXAMPLE

LEAP-002: dual primary endpoints with split alpha

Lenvatinib plus pembrolizumab versus lenvatinib plus placebo in advanced hepatocellular carcinoma

Overall survival

HR 0.84

95% CI 0.71–1.00

One-sided p = 0.023

Threshold p ≤ 0.019

Not met ✕

Progression-free survival

HR 0.87

95% CI 0.73–1.02

One-sided p = 0.047

Threshold p ≤ 0.002

Not met ✕

Design logic

- Two primary endpoints, each with its own α allocation and event-driven analysis
- Success on either endpoint would have supported the claim
- The PFS boundary was very strict after interim analyses, so nearly all α rested on OS
- Interpretation: “did not meet prespecified significance”; findings do not support a change in practice

Both hazard ratios favored the combination, yet neither crossed its prespecified boundary: a negative trial by design.

Llovet JM et al. Lancet Oncol 2023;24:1399–1410

Composite endpoints

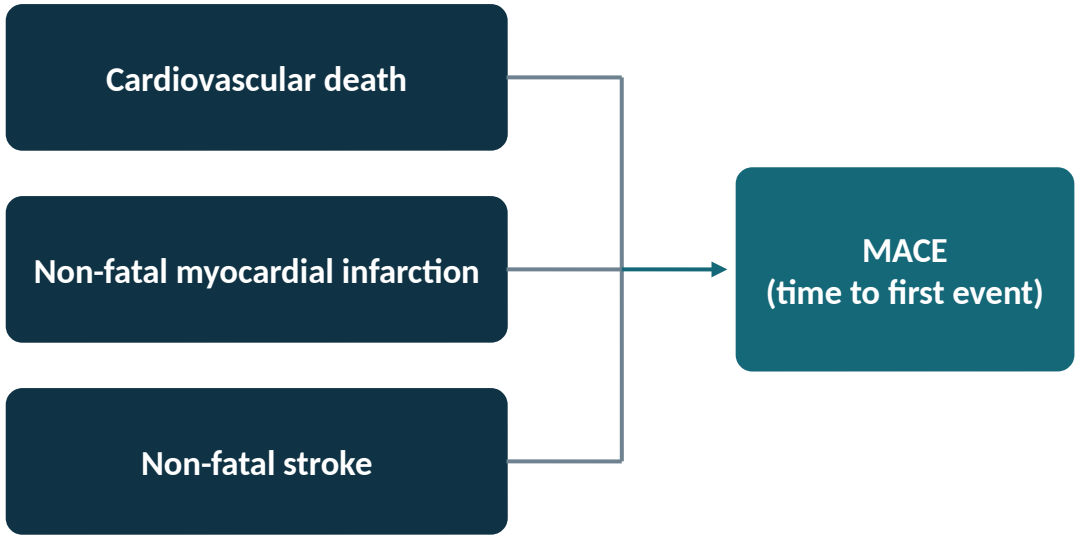
Several clinically related events combined into one outcome, usually time to the first of any

Why use one

- More events: smaller sample size and shorter follow-up
- One primary test: no multiplicity problem
- Captures the net clinical burden of a disease

Cautions

- Components should have similar clinical importance and respond similarly to treatment
- Often driven by the most frequent, least serious component
- Only the first event counts; later, worse events are ignored
- Can be difficult to interpret and to communicate to patients

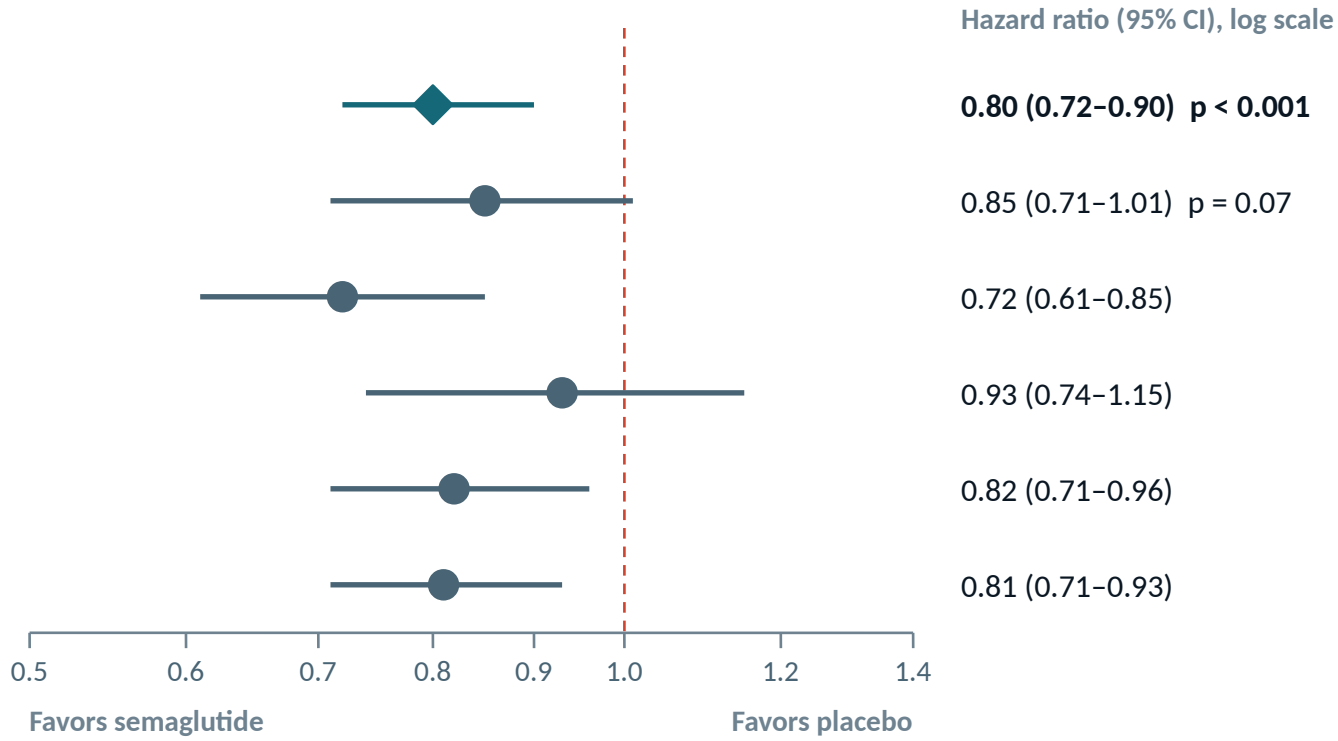


SELECT: 3-point major adverse cardiovascular events (MACE)

Always report each component: the composite tells you whether something happened, not what happened.

SELECT: reading a composite with its components

Semaglutide 2.4 mg versus placebo in overweight or obesity with cardiovascular disease; 17,604 patients, 40 months



The composite was clearly positive; the components show which events drove it and which are uncertain.

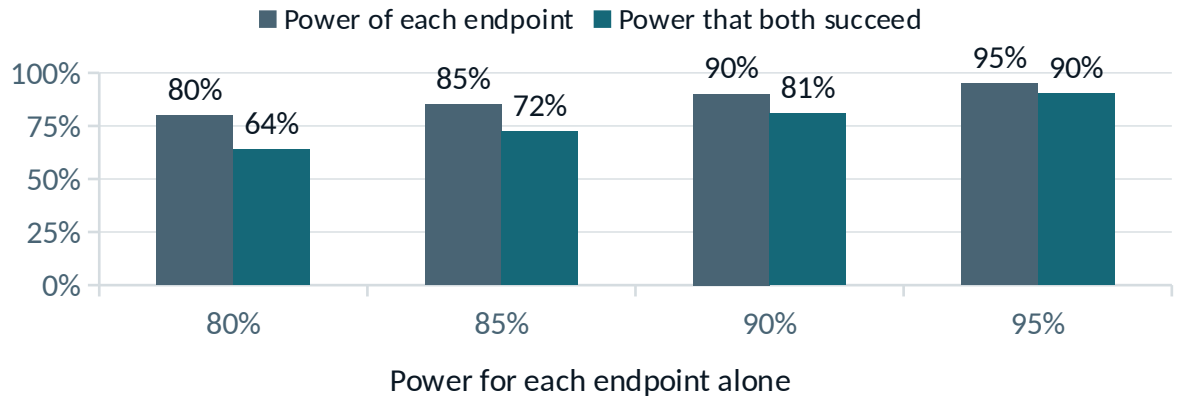
Lincoff AM et al. N Engl J Med 2023;389:2221–2232

Co-primary endpoints: every endpoint must succeed

Used when a treatment is clinically useful only if it works on two distinct domains

Endpoint A significant at α **AND** **Endpoint B significant at α** → **Trial succeeds**

Either fails → trial fails



Joint power ≈ product of the individual powers when endpoints are independent; higher when correlated.

Design implications

- Each endpoint tested at full α (no split): the “all must win” rule is already conservative
- Sample size must give adequate power for the pair, so it rises above the larger single requirement
- Typical settings: Alzheimer disease (cognition and function), IBS (pain and stool), MASH (histology)
- Different from multiple primary endpoints, where one success suffices and α is split

No alpha adjustment is needed, but the price is paid in power and sample size.

CLINICAL EXAMPLE

SMART: co-primary endpoints with different claims

Self-expanding (SEV) versus balloon-expandable (BEV) transcatheter aortic valves in small annuli; 716 patients, 12 months

Co-primary 1: clinical

Death, disabling stroke, or rehospitalization for heart failure

9.4% vs 10.6%

Difference −1.2 points (90% CI −4.9 to 2.5)

Non-inferiority: p < 0.001

Met ✓

Co-primary 2: valve function

Bioprosthetic-valve dysfunction

9.4% vs 41.6%

Difference −32.2 points (95% CI −38.7 to −25.6)

Superiority: p < 0.001

Met ✓

Why co-primary here

- A better-functioning valve is useful only if clinical outcomes are not worse
- The two domains answer different questions and use different hypotheses (NI versus superiority)
- Both tests run at their own α; success requires both
- Reporting states both results and both claims explicitly

One endpoint tested for non-inferiority, the other for superiority: both had to succeed, and both did.

Herrmann HC et al. N Engl J Med 2024;390:1959–1971

Choosing a strategy: match the claim to the method

What must the trial show?	Strategy	α handling	Power impact	Example
Benefit on at least one of several outcomes	Multiple primary endpoints with adjustment	Split or adjusted (Bonferroni, Holm, Hochberg)	Reduced per endpoint	LEAP-002: OS or PFS
Primary benefit, then ordered secondary claims	Hierarchical testing	Full α at each step, sequential	None for primary; later steps depend on earlier	Primary then key secondary endpoints
Net effect on related events	Composite endpoint	Single test	Increased (more events)	SELECT: 3-point MACE
Benefit on every domain	Co-primary endpoints	Full α for each; all must succeed	Reduced jointly; larger sample	SMART: clinical NI and valve superiority

Decide what the trial must show to change practice, then pick the testing rule that delivers exactly that claim.

GOOD PRACTICE

Aligning hypotheses and endpoints with clinical relevance

Patient-centered outcomes

Focus on what matters to patients: survival, symptoms, function, and quality of life. Involve patients in endpoint selection.

Disease-specific measures

Tailor the endpoint to the condition and its stage: histology in early MASH, decompensation events in cirrhosis.

Stakeholder input

Engage clinicians, patients, statisticians, and regulators early; use core outcome sets where they exist.

A statistically significant endpoint that patients and clinicians do not value will not change practice.

Good practice checklist

What reviewers, regulators, and journals will look for

Guideline compliance

ICH E9 and E9(R1), CONSORT, SPIRIT, and disease-specific regulatory guidance

Documentation

Hypotheses, endpoints, testing hierarchy, and sample size in the protocol and statistical analysis plan

Transparency

Registration before enrollment; any change to the primary endpoint dated, justified, and reported

Risk-benefit analysis

Endpoints capture harms as well as benefits; safety endpoints prespecified

Data integrity

Blinded assessment, independent adjudication, audit trail, missing-data strategy

Collaboration

Clinicians, statisticians, patients, and regulators involved from the design stage

Prespecify, register, document, and report: the credibility of the result rests on the paper trail.

APPLY THE CONCEPT

Class exercise: design the hypothesis and endpoints

Phase 3 trial of a new oral agent for MASH with stage F2–F3 fibrosis, 52 weeks, versus placebo

Scenario

- Regulators accept two histologic endpoints: MASH resolution without worsening of fibrosis, and fibrosis improvement ≥1 stage without worsening of MASH
- Sponsor also wants claims on non-invasive tests (liver stiffness, ALT) and a patient-reported fatigue score
- Paired biopsies are available for about 85% of patients

1. Write the primary hypothesis as H_0 and H_1 on the scale of the endpoint.
2. Should the two histologic endpoints be co-primary or multiple primary with split α? Justify.
3. Place the non-invasive and PRO endpoints in a testing hierarchy.
4. Which intercurrent event worries you most, and which estimand strategy would you choose?

Prepare a two-minute justification of your primary endpoint and your multiplicity plan.

Key principles

The hypothesis	Starts from a knowledge gap, is written as H_0 and H_1 on a measurable scale, and states its direction and margin.
The endpoint	Makes the hypothesis measurable; its data type determines the effect measure, the test, and the sample size.
The tiers	Primary carries the confirmatory claim; secondary endpoints support it; exploratory endpoints generate hypotheses.
Multiplicity	Every additional claim costs Type I error: adjust, order, combine, or require all, and decide before unblinding.
The practice	Clinical relevance, validated measurement, prespecification, registration, and transparent reporting.

A well-designed trial can be described in one sentence: the hypothesis, the primary endpoint, and the rule for success.

QUESTIONS AND DISCUSSION

Which endpoint would change your practice?

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